

Structure over Scale

Hierarchy-Constrained Modeling for English Online Polarization Detection
Mid-Submission Presentation

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The task: SemEval-2026 Task 9 (POLAR), English

- ▶ **det** (Subtask 1): is the text polarized?
- ▶ **type** (Subtask 2): which dimensions does it target?
Political, Racial/ethnic, Religious, Gender/sexual, Other
- ▶ The two are **nested**: types exist only for polarized texts

$$d = 1 \iff \bigvee_{k=1}^5 t_k = 1$$

- ▶ The gold labels satisfy this rule with **zero** exceptions

English training data

Texts	3,222
Polarized / not	1,175 / 2,047
Political	1,150
Racial/ethnic	281
Religious	112
Gender/sexual	72
Other	126
≥2 types	426

Political is on 98% of polarized texts;
Gender and Religious are rare.

The problem and prior work

Systems break the hierarchy

- ▶ Separate predictions allow “not polarized” with a type (LVR-A) or “polarized” with no type (LVR-B)
- ▶ POLAR teams fix this *after* prediction:
 - ▶ YEZE: independent modelling “causes hierarchical violations”
 - ▶ NYCU-NLP: polarized but no type → *Other*
 - ▶ Sagarmatha: $\hat{d} = 0 \Rightarrow$ no types; “slightly limits raw multi-label recall”

Gap and question

No POLAR system makes DET a *function* of the types, or compares post-hoc and training-time consistency under one controlled setup. **Should the hierarchy be part of the model?**

The landscape

- ▶ 67 teams; LLMs dominate (Qwen 31%, LLaMA 20%, Gemma 19%; encoders 7%)
- ▶ English best: DET 0.825, TYPE 0.532 (12B–27B models)
- ▶ Encoders stay competitive with tuned thresholds and multi-task learning

Research questions

RQ1

Does deriving DET from TYPE (noisy-OR) improve DET and TYPE macro-F1 over independent models and multi-task learning?

H1: M4-core $>$ M1 and M2 on both metrics (4 tests)

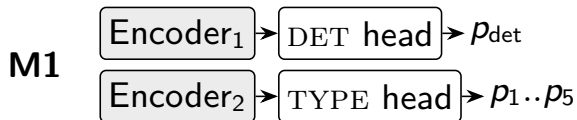
RQ2

At equal consistency, does building the hierarchy into training keep more TYPE recall than fixing outputs afterwards?

H2: M4-core $>$ M3 on TYPE macro-F1 and macro-recall (2 tests)

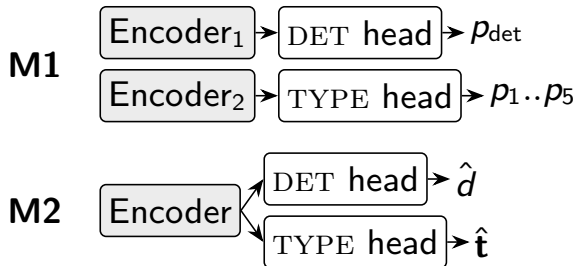
The six tests were fixed before any experiment and are Holm-corrected together. The “scale” comparison with 12B–27B systems is a final-stage deliverable (RQ4).

Four models, one encoder (DeBERTa-v3-base, 184M)



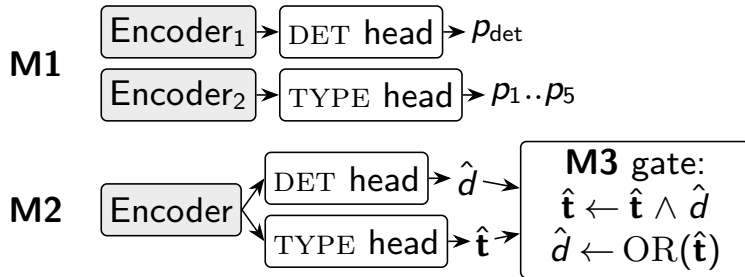
M1: two independent encoders, one per task (368M parameters).

Four models, one encoder (DeBERTa-v3-base, 184M)



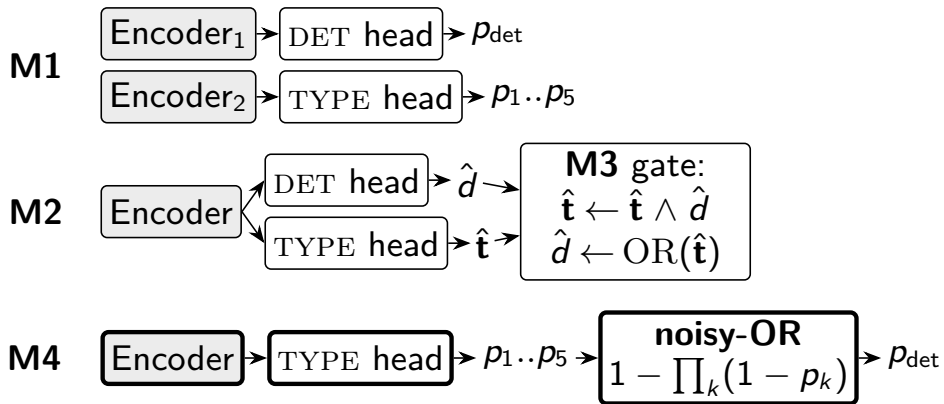
M2: one shared encoder, two unconnected heads.

Four models, one encoder (DeBERTa-v3-base, 184M)



M3: M2's predictions repaired after decoding; no training.

Four models, one encoder (DeBERTa-v3-base, 184M)



M4-core: no DET head, detection is computed from the types: $p_{\text{det}} = 1 - \prod_k (1 - p_k)$.

M4-core: detection from the types

Training

$$p_{\text{det}} = 1 - \prod_{k=1}^5 (1 - p_k)$$

$$\mathcal{L} = \text{BCE}(y_{\text{det}}, p_{\text{det}}) + \lambda \mathcal{L}_{\text{type}}$$

TYPE loss only on polarized texts
(types are undefined elsewhere)

Inference

$$\hat{t}_k = \mathbf{1}[p_k \geq \tau_k], \quad \hat{d} = \bigvee_k \hat{t}_k$$

Contradictions are impossible:
LVR = 0 by construction

Why it should help

On a *non-polarized* text, the DET loss pushes *every* type probability down, so the type head is supervised exactly where the TYPE loss is masked.

Experimental setup

Protocol

- ▶ 5 fixed stratified folds $\times$ 5 seeds
- ▶ 4 trained models $\rightarrow$ **100 training runs**
- ▶ 10% inner validation: early stopping, checkpoint and thresholds
- ▶ Each held-out fold predicted once
- ▶ 3.9 GPU-hours on one RTX 4060

Metrics and statistics

- ▶ DET macro-F1 (both classes)
- ▶ TYPE macro-P/R/F1 on *gold-polarized* texts (pre-declared)
- ▶ TYPE on *all* texts (exploratory)
- ▶ LVR-A / LVR-B (violations)
- ▶ Paired bootstrap (2,000), Holm over 6 tests

Thresholds (key for the results)

TYPE thresholds are tuned on **polarized** inner-validation texts only. M1 and M2 also tune a DET threshold; **M4-core has no det threshold**.

Main results

Model	DET	TYPE, polarized texts			TYPE, all texts	Violations
	macro-F1	macro-F1	macro-P	macro-R	macro-F1	LVR
M1 independent	0.799	0.543	0.509	0.600	0.299	61.9%
M2 shared MTL	0.793	0.497	0.444	0.595	0.255	61.6%
M3 gating	0.793	0.453	0.469	0.468	0.382	0.0%
M4-core	0.784	0.461	0.465	0.474	0.396	0.0%

- ▶ Out-of-fold over 3,222 texts; mean over 5 seeds
- ▶ M1 is best on every pre-declared metric
- ▶ Consistent models (M3, M4) lose on TYPE **recall** (0.47 vs. 0.60), not precision (0.47 vs. 0.44–0.51)

Consistency: unconstrained models contradict themselves

62%

of all texts violate the hierarchy
(M1 and M2, almost all LVR-A)

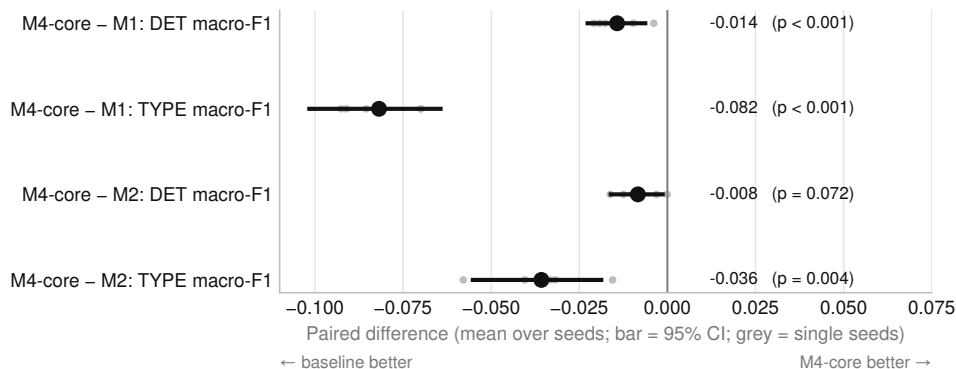
0%

for M3 and M4-core, in every run

Why so high?

- ▶ M2 gives a type to *all* 2,047 neutral texts
- ▶ The TYPE loss never sees a neutral text
- ▶ Thresholds are tuned only on polarized texts
- ▶ Nothing teaches the type heads to stay silent

RQ1: noisy-OR does not beat M1 or M2



H1 not supported. Three of four differences are significant after Holm correction, all favouring the baselines; no seed favours M4-core.

RQ1: why does M4-core lose on DET?

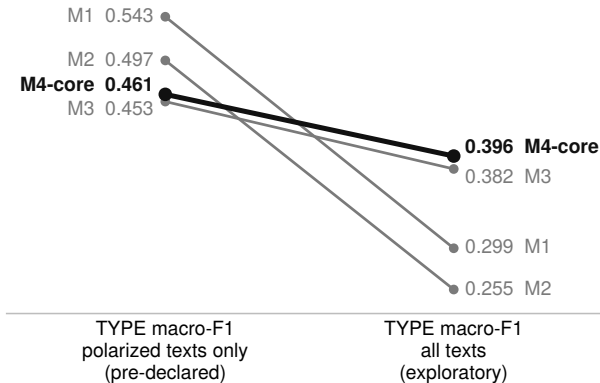
	Recall	Precision	False alarms
M2	0.761	0.722	345
M4-core	0.831	0.674	472

per seed, over 3,222 texts

- ▶ M4-core finds **82 more** polarized texts. . .
- ▶ . . . but flags **127 more** neutral ones
- ▶ It says “polarized” if *any* type fires
- ▶ Its type thresholds never saw a neutral text
- ▶ The Political threshold hit the grid floor (0.20) in 19 of 25 runs

The DET comparison is not calibration-matched: M1 and M2 tune a dedicated DET threshold, M4-core cannot.

The scoring scope decides the winner



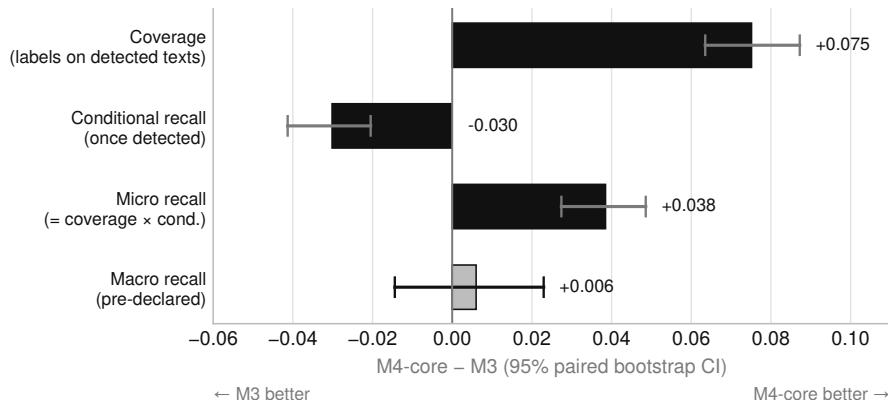
- ▶ Pre-declared metric ignores types predicted on neutral texts
- ▶ A consistent model can only type what it detects
- ▶ Scored on all texts, the ranking **reverses**

Exploratory

Not part of the pre-declared tests. The official scorer code is not public, so its scope is unverified; the official data store all-zero type labels for non-polarized texts, which is consistent with all-text scoring.

RQ2: gating and noisy-OR tie, because two effects cancel

H2 not supported: TYPE F1 +0.008 [−0.009, +0.022]; macro-recall +0.006 [−0.014, +0.023]; Holm $p = 0.73$.



For a consistent model, label recall = coverage × conditional recall. M4-core detects more; M3 discriminates better once detected. Decomposition is exploratory.

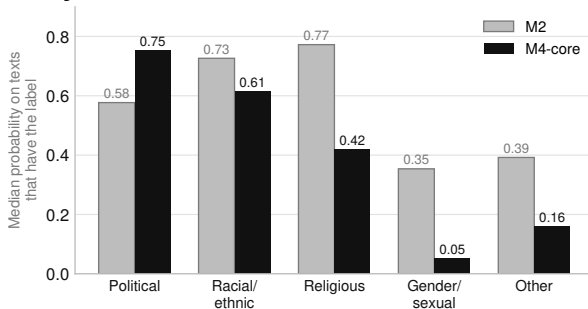
RQ2 per label: Political gains, rare labels lose

Label	<i>n</i>	Δ recall [95% CI]	Δ F1 [95% CI]
Political	1,150	+0.067 [+0.055, +0.078]	+0.040 [+0.033, +0.048]
Racial/ethnic	281	-0.011 [-0.041, +0.019]	-0.026 [-0.051, -0.003]
Religious	112	-0.016 [-0.059, +0.026]	-0.006 [-0.041, +0.030]
Gender/sexual	72	+0.064 [+0.009, +0.116]	+0.039 [-0.014, +0.089]
Other	126	-0.073 [-0.113, -0.034]	-0.009 [-0.039, +0.019]

- ▶ M4-core – M3; bold = interval excludes zero (exploratory)
- ▶ The coverage gain lands on Political, which carries two thirds of all labels
- ▶ Macro averaging weights every label equally, so gains and losses cancel

RQ2: why rare labels lose, and what gating costs

Noisy-OR concentrates credit in Political



Political alone can explain “polarized”, so rare heads are pushed down even on texts that truly have them.

Gating deletes correct labels

- ▶ Per seed: 4,379 of 6,647 labels removed
- ▶ 87.7% were false alarms on neutral texts
- ▶ But **338 correct labels** (22.9%) were removed too
- ▶ On 98.7% of missed polarized texts, M2 already had a correct type

Error propagation from the parent decision.

Takeaways

1. **Consistency has a built-in cost.**

A consistent model cannot type a text it fails to detect. Where the constraint is enforced changes who pays, not whether.

2. **The metric decides the winner.**

Scoring TYPE on polarized texts vs. all texts reverses the ranking.

3. **Calibration is part of the model.**

M4-core's main weaknesses come from its thresholds and the direction of the hierarchy, both testable.

Verdicts stay frozen: RQ1 not supported, RQ2 inconclusive.

Limitations

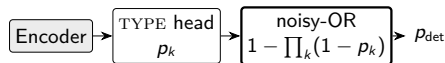
- ▶ **det is not calibration-matched.** M1 and M2 tune a DET threshold; M4-core cannot, and M1-DET also selects its checkpoint on DET.
- ▶ **M1 uses twice the parameters** (368M vs. 184M).
- ▶ **Some analyses were designed after seeing the results:** all-text scoring, the coverage decomposition, the probability profile and per-label differences are exploratory and uncorrected.
- ▶ **RQ4 splits differ.** Published 12B–27B scores are on the official test set; ours are cross-validation on the training set. The official test labels are public (identical to our copy), so the final stage can close this gap.
- ▶ English only, five seeds, one GPU; official scorer scope unverified.

Remaining work, in order

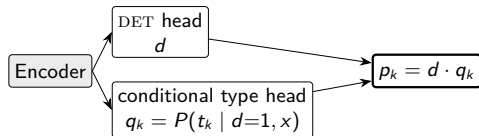
1. **Decoder extension (first).** Tune thresholds on *all* inner-validation texts (grid 0.05–0.80); rerun M4-core and M2; compare M3 and M4-core at the same detection operating point.
Why: the cheapest test of whether RQ1/RQ2 are decoding artefacts.
2. **RQ5 (new): parent-to-child model M5.**
Why: a confident type should not be able to switch detection on.
3. **RQ3 (proposal): label-aware heads** and a separate Other head, on the variant kept after RQ5.
Why: rare labels and Other are weakest (F1 below 0.30).
4. **RQ4 (proposal): 184M vs. 12B–27B**, beside the published results; no leaderboard claim.
Why: the “structure over scale” question.

RQ5 hypothesis: reverse the direction of the hierarchy

M4-core: types $\rightarrow$ detection



M5: detection $\rightarrow$ types



- ▶ In M5, low detection confidence softly lowers every type, and a confident type can no longer switch detection on; soft and hard decoders.
- ▶ **H5a:** M5 beats M2 and the decoder-extension M4-core on TYPE macro-F1.
- ▶ **H5b (non-inferiority):** DET macro-F1 within $\delta = 0.01$ of M2, decided by the *lower end* of the 95% interval.

A hypothesis to test after the decoder extension, not a result.

Timeline to the final submission (Oct 31)

Dates	Work
Oct 2–3	Read the official scorer; record the TYPE scoring scope
Oct 4–6	Decoder extension: rerun M4-core and M2 (25 runs each)
Oct 7–16	RQ5: parent-to-child M5, soft and hard decoders
Oct 17–23	RQ3: label-aware heads on the kept variant
Oct 24–27	RQ4: comparison with published 12B–27B results
Oct 28–31	Error analysis, final report and code release

Each 25-run condition takes about one GPU-hour: compute is not the risk.

Thank you

Questions?

Code https://github.com/Rakshitagg06/ANLP_Ugh

Runs <https://wandb.ai/rakshitagg06-iiit-hyderabad/anlp-project>

Backup: why not let the types switch detection on?

Q: M2 already had a correct type on 98.7% of the polarized texts its detector missed. Why not set $\hat{d} = \text{OR}(\text{types})$ instead of gating?

A: Because M2's type head fires on **every** neutral text.

- ▶ M2 assigns at least one type to all 2,047 of 2,047 neutral texts
- ▶ $\hat{d} = \text{OR}(\text{types})$ would therefore call every text polarized
- ▶ M4-core is the *trained* version of this idea, and it still raises 127 more false alarms per seed than M2
- ▶ That is why RQ5 reverses the direction: detection should gate the types, not the other way round

Backup: gating audit (M2 $\rightarrow$ M3, per seed)

Quantity	Mean	Share
Type labels predicted by M2	6,647	
Labels removed by gating	4,379	
wrong, on neutral texts	3,839	87.7%
wrong, on polarized texts	202	4.6%
correct , on polarized texts	338	7.7%
Polarized texts gated (M2 detection misses)	281	
with ≥ 1 correct M2 type	277	98.7%

Recall cost of gating vs. M2: -0.127 macro-recall; larger on single-label texts (-0.239) than on multi-label texts (-0.161).

Backup: progress against the proposal

Proposal item	Status
Stratified 5-fold CV, metrics, LVR	Done
M1 and M2 (5 folds $\times$ 5 seeds)	Done
M3 gating	Done (re-implemented)
M4 noisy-OR	Done as M4-core
Full ladder and paired tests	Done (ahead of plan)
Label-aware attention, Other head	Kept as RQ3
TF-IDF, BERT, RoBERTa baselines	Deferred
Comparison with 12B–27B systems	Kept as RQ4